

# Leaf Convergence Games: Partizan Games on Terminating Rooted Graphs

IE619 | Combinatorial Game Theory (2025)

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# Abstract

We introduce **Leaf Convergence Games (LCG)** — partizan games on finite rooted DAGs where **every maximal path terminates at a designated leaf**.

Left moves on blue (**L**) edges, Right on red (**R**). Normal play.

**LCG provides a clean graph-theoretic lens** on classical CGT values (numbers, switches, infinitesimals, dyadics) while guaranteeing termination without surreal numbers or loopy machinery.

# Why LCG? — The “So What?”

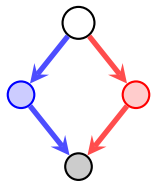
**Standard CGT** studies abstract game values. **LCG** gives a **canonical structural model**.

## Key Contributions:

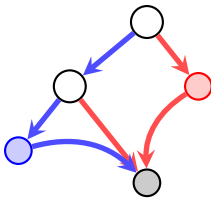
- Guaranteed termination by construction (no cycles, no infinite plays).
- Values **emerge directly from graph topology**.
- Bridges graph theory and CGT — a natural model for short partizan games.
- Every short partizan game has an LCG representation (Representation Theorem).

**LCG = Where CGT values meet concrete graph structure.**

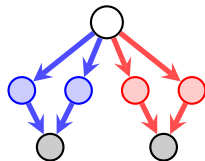
# Motivating Structures



Linear



Mixed depth



Multi-sink

# Value Classification in LCG

## Theorem (Numbers)

*A position is a number  $\iff$  every Left option  $<$  every Right option.*

## Theorem (Integers)

*Integers correspond exactly to pure directed chains (unbranched Left/Right paths).*

## Theorem (Switches)

*A position is a switch  $\iff \exists$  Left option  $> 0$  and  $\exists$  Right option  $< 0$ .*

## Theorem (Infinitesimals)

*Infinitesimal  $\iff$  all Left options  $\leq 0$ , all Right options  $\geq 0$ , but not all zero.*

These characterisations link **graph topology** directly to classical CGT value types.

## Important Clarification: Branching $\neq$ Magnitude

**Common misconception:** More Left edges  $\implies$  bigger positive number.

**Fact:** Value depends on the **values of options**, not their count.

- Duplicate identical options do not change the game value.
- Branching creates choice, not raw strength.
- True magnitude comes from depth and asymmetry in subtrees.

Graph topology  $\rightsquigarrow$  strategic options, not raw count.

## How to Read an LCG Graph — Visual Intuition

- **Long Left-only path**  $\rightarrow$  large positive integer
- **Long Right-only path**  $\rightarrow$  large negative integer
- **Balanced symmetric split**  $\rightarrow *$
- **Slight Left advantage at depth**  $\rightarrow$  positive infinitesimal ( $\uparrow$ )
- **Competing advantages**  $\rightarrow$  switch ( $\pm n$ )

**LCG makes CGT values visually readable from the graph structure.**

# Formal Framework

**LCG Position:** Finite rooted DAG + terminal leaves + L/R edge labels + convergence property.

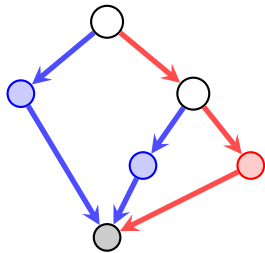
**Value Definition:**

$$\mathcal{G}(v) = \{\mathcal{G}(c) : c \in \mathcal{L}(v) \mid \mathcal{G}(c) : c \in \mathcal{R}(v)\}$$

with  $\mathcal{G}(\tau) = 0$  for terminals.

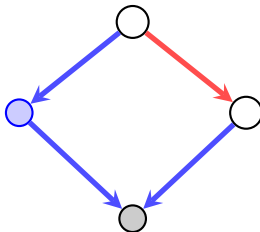


## Example 1: Asymmetric Depth Tree



Value:  $\{0 \mid *\} = \uparrow$

## Surprising Example: Value = $1/2$



**LCG can realise all dyadic rationals** through careful asymmetric branching, not obvious from the graph at first glance.

# Computational Analysis

## Python evaluator:

```
def evaluate_lcg(nodes, edges, root, terminals):
    memo = {}
    def eval_node(v):
        if v in terminals: return ([], [])
        left_opts = [eval_node(c) for s,c,col in edges if s==v and col
                        == 'L']
        right_opts = [eval_node(c) for s,c,col in edges if s==v and
                        col=='R']
        return (left_opts, right_opts)
    return eval_node(root)
```

## Relation to Existing Frameworks

- **Game Trees (standard CGT)** — LCG is a restricted, terminating subclass.
- **Surreal Numbers** — LCG realises the same values without needing birthdays.
- **Hackenbush** — Similar visual intuition, but on general graphs.
- **Representation Theorem (informal)**: Every short partizan game has an equivalent LCG.

# Key Takeaways

- 1 Graph topology.
- 2 Strong classification theorems link topology  $\rightarrow$  value type.
- 3 Branching adds nuance.
- 4 Rich values (integers, switches, infinitesimals, dyadics) appear naturally.

## Open Questions & Future Work

- ① Efficient canonical reduction algorithm for arbitrary LCG DAGs.
- ② Complete enumeration and growth rate of values at depth  $d$ .
- ③ Algorithmic complexity of optimal play in LCG.
- ④ Generalisation to loopy or fuzzy variants.

**LCG opens a graph-theoretic pathway into classical Combinatorial Game Theory.**

# References

-  Berlekamp, Conway, Guy, *Winning Ways*.
-  Conway, *On Numbers and Games*.
-  Albert et al., *Lessons in Play*.
-  Siegel, *Combinatorial Game Theory*.
-  CGSuite.